Fiscal Year:	FY 2012	**Task Last Updated:**	FY 12/12/2011
PI Name:	Bungo, Michael W. M.D.		
Project Title:	Cardiac Atrophy and Diastolic Dysfunction During and After Long Duration Spaceflight: Functional Consequences for Orthostatic Intolerance, Exercise Capacity, and Risk of Cardiac Arrhythmias		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	02/01/2004	**End Date:**	03/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Baumann, David	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS Increment 19 (info from PI 2/2009); ISS 19-25 (Ed., 12/12/2011) NOTE: End date is 3/31/2014 per JSC HRP Master Task List update 5/20/2011 (Ed., 7/21/2011) NOTE: End date corrected to 9/30/2012 per J. Dardano/JSC and PI; previous end date showed 1/10/2009 in error (1/2009 information)		
Key Personnel Changes/Previous PI:	Co-PI: Benjamin D. Levine, M.D. ; Co-PI: Michael W. Bungo, M.D.		
COI Name (Institution):	Levine, Benjamin (UT - Southwestern)		
Grant/Contract No.:	NNJ04HB03A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cardiac atrophy appears to develop during spaceflight or its ground-based analogues, leading to diastolic dysfunction and orthostatic hypotension. Such atrophy also may be a potential mechanism for the cardiac arrhythmias recently identified in some crew members after long duration exposure to microgravity aboard the Mir space station. Recent work has suggested that cardiac atrophy may be progressive, without a clear plateau over at least 12 weeks of bed rest, and thus may be a significant limiting factor for extended duration space missions. The purpose of this experiment is to quantify the extent, time course, and clinical significance of cardiac atrophy and identify its mechanisms. The functional consequences of this atrophy also will be determined for cardiac filling dynamics, orthostatic tolerance at 1G and fractional G (Mars and Moon) conditions, exercise tolerance, and arrhythmia susceptibility both in space on the ISS, and following return to earth. Three specific aims will be addressed: Objective 1a: To determine the magnitude of left and right ventricular atrophy associated with long duration space flight, and to relate this atrophy to measures of physical activity and cardiac work in-flight. Magnetic resonance imaging (MRI) will be performed pre- and post-flight as the most accurate means of measuring cardiac mass, including sufficient repetitions to document recovery. Objective 1b: To determine the time course and pattern of progression of cardiac atrophy in-flight using cardiac ultrasound. Objective 2: To determine the functional importance of cardiac atrophy for cardiac diastolic function and the regulation of stroke volume during gravitational transitions, by using innovative, non-invasive imaging techniques to measure the dynamic component of diastole including: a) Strain Rate and Tissue Doppler Imaging (TDI) to quantify ventricular wall motion with high temporal resolution; b) Color Doppler M-mode echocardiography to quantify intraventricular pressure gradients and assess the magnitude of diastolic suction; and c) MRI with myocardial tagging to quantify the rate of untwisting. Objective 3: To identify changes in ventricular conduction, depolarization and repolarization during and after long duration spaceflight, and relate these to changes in cardiac mass and morphology (presence of fibrosis from late gadolinium enhancement and/or fat accumulation from MR spectroscopy). Although the direct assessment of arrhythmia risk in otherwise healthy subjects is imprecise, the goal of this objective is to combine the best currently available non-invasive tests of arrhythmia risk with high negative predictive value to demonstrate convincingly the absence of risk for life threaten arrhythmias. These tests will include quantifying: 1) Arrhythmia burden; 2) Delayed conduction via the signal averaged electrocardiogram (SAECG); 3) Dispersion of refractoriness from QT dispersion (QTD), QT variability and T-loop morphology; and 4) Microvolt T-wave alternans (TWA) during daily activity including routine exercise. After completion of this study, the clinical expression of cardiac atrophy during long duration space flight will be defined clearly, and its significance for cardiac systolic and diastolic function at rest and during gravitational transitions will be elucidated. In addition, preliminary information will be obtained regarding ventricular conduction and repolarization that will provide either strong clinical reassurance, or pathophysiologic insight into the risk for cardiac arrhythmias. Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained from these spaceflight experiments also will be relevant for patients after prolonged confinement to bed rest, as well as conditions that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.
Task Progress:	To date, six crewmembers have completed the full protocol, a seventh is aboard the ISS, and two additional are in the last month of preflight data collection. The calendar year 2010 included the major failure of the ISS ultrasound machine during a medical operations scan in January. The device was deemed un-repairable and the new “Ultrasound 2” replacement unit was delivered by the Shuttle in June of the year. Except for early inflight data, this failure resulted in the loss of all flight echocardiography on subject number five and loss of data prior to FD60 on subject number 6. The Cardiopress equipment also failed during June of the year and the equipment was returned via the same Shuttle flight. A replacement unit is scheduled to be delivered to the ISS in late December 2010. Final impact of these anomalies will not be sufficiently defined until all data is collected and analyzed, since the impact of biologic variability versus temporal variability will not be defined until those data are available; however, at this time, with a thirteenth subject planned to be concluded by June 2013, it is not expected that these anomalies will have critical impact upon the ability to derive sufficient conclusions from data analyses. As the limited discussion presented above details, each technique contributes to several objects and each objective can be at least partially achieved by several of the techniques. This design allows for significant data “usability”. Thus, deviations from the initially proposed schedule and the initially proposed techniques will have impact upon the fidelity of conclusions that may result from the project, but the overall impact may be mitigated by the successful completion of other components of the study and protection against “catastrophic” failure ensured. Data analysis is ongoing. We are reluctant to analyze too much data prior to completion of all studies by an individual astronaut so that we can sustain blinded analysis and avoid bias. Nevertheless, each data set (echo imaging, MR imaging, rhythm and BP monitoring) have been analyzed in part to determine data quality. Initial data analyses has resulted in presentations of partial pooled data sets at professional meetings. These “abstracts” and “posters” are referenced in the Task Book Bibliography section.
Bibliography Type:	Description: (Last Updated: 01/12/2021)

Abstracts for Journals and Proceedings	Shibata S, Hastings JL, Gandhi N, Brian J, Liang D, Platts S, Bungo MW, Levine BD. "Cardiac Work and Mass During Long Duration Spaceflight." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Hastings JL, Daniels J, Shibata S, Platts S, Hamilton D, Page R, Goldberger J, Bungo MW, Levine BD. "Cardiac Arrhythmias During Long Duration Spaceflight." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011